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Primary Care Physician–led Population-based Screening Program for Prostate Cancer: Bridging a Urologic Gap in Access to Care

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A B S T R A C T

Objective: To evaluate the feasibility and outcomes of a primary care–led, community-based prostate-specific antigen (PSA) screening program designed to improve access to diagnosis in underserved populations.

Methods: We retrospectively analyzed two cohorts: a Primary Analysis Cohort (n = 927; 6/2022-8/2023) to assess guideline adherence and clinical outcomes, and an Extended Programmatic Cohort (n = 1773; through 12/2024) to evaluate program scalability. Per 2023 AUA guidelines, screening was "appropriate" for men aged 45-69, or aged 40-44 with high-risk features (Black ancestry or family history). Patients with PSA > 4 ng/mL underwent repeat testing and/or mpMRI triage.

Results: In the primary cohort, 833 (89.8%) participants were screened appropriately. Of the 94 (10.2%) screened outside guidelines, none were referred to urology. Among appropriately screened men with PSA > 4 ng/mL (n = 71), 49 underwent mpMRI, with 23 (47%) demonstrating PI-RADS 3-5 findings. Twelve patients underwent biopsy, yielding nine (75%) prostate cancer diagnoses. Notably, 26 patients (48% of those with elevated PSA and completed work-ups) safely avoided urology referral through negative MRI findings or normalized repeat PSA results. In the extended cohort (n = 1773), the program resulted in 50 total urology referrals and 21 patients receiving definitive treatment.

Conclusion: This primary care–driven model effectively identifies clinically significant prostate cancer while maintaining high (90%) guideline adherence and reducing unnecessary specialist referrals. By integrating mpMRI triage, this framework provides a scalable strategy to bridge urologic care gaps and mitigate health disparities in underserved communities.

Prostate cancer is the most common malignancy affecting men, with 1 in 8 estimated to be diagnosed in their lifetime. The American Cancer Society estimates about 299,010 new cases of prostate cancer and 35,250 deaths in 2024.[1,2] The advent of prostate specific antigen (PSA) screening led to a marked reduction in prostate cancer specific mortality associated with a downward stage migration resulting from earlier detection.[3] The screening paradigm shifted in the 2000s, marked by the grade D recommendation against screening with PSA given by the US Preventive Services Task Force (USPSTF) in May 2012.[4,5] This recommendation, which incorporated findings from trials such as the Prostate, Lung, Colorectal Cancer Screening Trial (PLCO), differed from the American Urologic Association (AUA) guidelines and the practice patterns of Urologists until May of 2018 when the USPSTF updated the recommendation to grade C for screening in men aged 55 to 69 years.[6]

The shift in screening in the 2010s was followed by decreased rates of screening and a migration toward higher grade and stage of cancer.[7] Prior to the 2012 USPSTF decision, Black men experienced a 134% higher mortality rate from the disease compared to non-Hispanic white counterparts.[8] In the post-USPSTF era (May 2012), research has noted a decrease in the survival disparity decreased between Black and non-Hispanic White men attributed to decreased survival of non-Hispanic White men and not an improved survival in black men.[9] Access to PSA screening in this high-risk population has remained a challenge, one that is exacerbated by access to urologic specialty care.[10-14] Numerous studies have reflected concerns over the gap between the need for urologic expertise and availability of providers, especially in rural communities and underserved populations.[15-17]

While clinical practice guidelines have encouraged the implementation of population-based PSA screening (European Association of Urology, AUA) there remains a gap in care for populations at high risk.[18] The AUA guidelines for the early detection of prostate cancer prioritize a risk stratified approach, suggesting that men at higher risk, including Black men and those with a significant family history, engage in shared decision making as early as age 40 to 45. This urological strategy seeks to identify high grade disease in younger patients who stand to gain the most from early curative treatment. Efforts to bridge the gap include shifting the role of screening and diagnostic work-up to primary care physicians and community-focused efforts to bring

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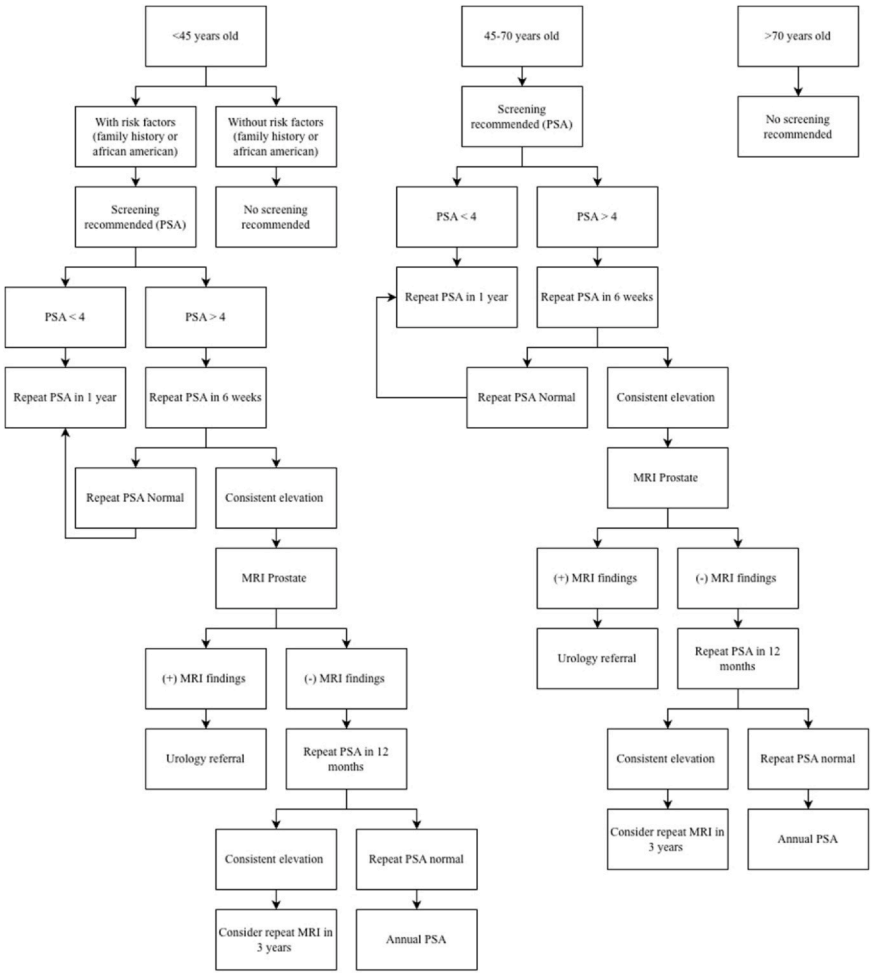


Figure 1. Updated MCI prostate cancer screening algorithm.

screening to vulnerable populations.[19-21] Herein, we present an early detection program for prostate cancer in Tarrant County, Texas, focusing on the delivery of high-quality care to vulnerable populations, including those with low incomes, limited access to physicians, and inadequate insurance coverage. We also discuss the feasibility of shifting the diagnosis, work-up, and coordination of care from the urologist to primary care physicians and community programs.

The objective of this study was to evaluate the feasibility of a primary care led prostate cancer screening program in reaching underserved populations. Furthermore, recognizing the clinical complexity of screening at the extremes of the recommended age range, we conducted targeted sub-analyses for patients aged < 45 and > 70. These secondary investigations aimed to evaluate the utility of our triage model in navigating the unique risks of early-onset disease and overdiagnosis, respectively. We hypothesized that a structured, multidisciplinary triage pathway utilizing repeat PSA testing and mpMRI could achieve high adherence to national screening guidelines while safely reducing the number of unnecessary urology referrals.

MATERIALS AND METHODS

The Moncrief Cancer Institute (MCI) is a non-profit, community-focused center in Fort Worth, Texas dedicated to early detection and cancer support services. The institute is part of the UT Southwestern Harold C. Simmons Comprehensive Cancer Center and offers screening programs for breast, cervical, colorectal, lung and prostate cancer, educational sessions for patients with cancer diagnoses, survivorship programs, nutrition and fitness programs. The American Rescue Plan Act (ARPA) provided funds in 2021 to support cancer screening programs in Tarrant County with the primary goal of increasing screening and reducing the estimated delays in care secondary to COVID 19. The MCI supported prostate cancer screening program starting in June of 2022 with the primary goal to serve uninsured and underinsured patients in Tarrant County. The program structure centers on community screening events and individual appointments at MCI. The primary screening events occur at either the MCI or via mobile screening clinics at health fairs and community events. Patients eligible for screening include low-income, uninsured or underinsured residents who present for screening at MCI or various screening events. There was no age cut off for screening at the time of program initiation.

After obtaining a PSA, a patient navigator assists in coordinating the follow-up for the patient. The navigators coordinate care via a centralized toll-free number with a bilingual team who assist with scheduling. For patients with a normal PSA, they are invited to return to future screening events at appropriate intervals and their clinical and contact information is gathered to ensure appropriate follow-up. For patients with an elevated PSA > 4 ng/mL, they are recommended to obtain a repeat PSA. Pending the results of the confirmatory PSA, patients with persistent elevations obtain a multi-parametric magnetic resonsonance image (MRI) of the prostate (mpMRI Prostate) performed at the MCI. Based on results, patients are triaged to repeat PSA screening or are referred to the Department of Urology at John Peter Smith Hospital (JPS; Tarrant County) or Parkland Health and Hospital System (PHHS; Dallas County). The decision for future diagnostics or referrals is performed by a Family Medicine trained physician who runs the program. Patients who are diagnosed with clinically significant

Table 1
PSA results, mpMRI results and clinical outcomes of the research period (6/1/22-8/31/23).

	Total Cohort (N = 927)
Demographics	
Age (y), median [IQR]	53 [48-59]
Race, n (%)	
White	608 (65.6%)
Black or African American	126 (13.6%)
Other or Unknown	193 (20.8%)
Ethnicity, n (%)	
Hispanic or Latino	686 (74.0%)
Non-Hispanic or Latino	209 (22.5%)
Unknown or Other	32 (3.5%)
Clinical variables	
Median PSA level (ng/mL) [IQR]	0.81 [0.47-1.48]
PSA type (total), n (%)	927 (100%)
Family history of prostate cancer (yes), n (%)	112 (12.1%)
Urologic symptoms (LUTS), n (%)	303 (32.7%)
Elevated PSA > 4 ng/mL	71 (7.7%)
Repeat PSA	34 (3.7%)
mpMRI performed	49 (5.3%)
PI-RADS 1	15 (1.6%)
PI-RADS 2	11 (1.2%)
PI-RADS 3	8 (0.9%)
PI-RADS 4	5 (0.5%)
PI-RADS 5	10 (1.1%)
Referrals to Urology	28 (3.0%)
Avoided referral to Urology	26 (2.8%)
Prostate biopsies	12 (1.3%)
Positive findings on biopsy	9 (1.0%)
Treatment for prostate cancer	8* (0.9%)
Active surveillance	3 (0.3%)
RALP	2 (0.2%)
XRT	2 (0.2%)
Metastatic disease treatment	1 (0.1%)

LUTS, lower urinary tract symptoms; PSA, prostate-specific antigen; RALP, robotic assisted laparoscopic prostatectomy; XRT, external beam radiation therapy.
* One patient was lost to follow up after diagnosis of prostate cancer.

prostate cancer are navigated through appropriate treatment with the assistance of the clinical navigators. The funds provided by the ARPA support the appointments, diagnostic testing and treatment for these patients. This process is centralized at MCI, which serves as the financial hub, providing reimbursement based on contracted Medicare rates.

After 2 years of the program (6/1/22 to 8/31/23), the MCI engaged with physicians at the Department of Urology at JPS to evaluate the programs' adherence to guidelines and continue to build on the working relationship between MCI and the department. We therefore sought to evaluate adherence of the screening protocol in accordance with the American Urology Association (AUA) Guidelines on prostate cancer screening. Data from the initiation of the screening protocol to 8/31/23 was obtained, including patient age, ethnicity, urinary symptoms, and family history. Data on subsequent visits, including repeat PSA, mpMRI Prostate, biopsy results and recommendations were obtained. The screening protocol was evaluated against the AUA Guidelines and Urology provider assessment. The AUA Guidelines for Early Detection of Prostate Cancer (2023) were used. This information is presented below and was used in meetings with the MCI to create a standardized follow-up program and review the challenges of a large population-based screening protocol with guideline adherence (Fig. 1).

RESULTS

Review Period (June 1, 2022-August 31, 2023)

During the original review period, a total of 927 patients were screened with a PSA and 71 had a PSA elevated > 4 ng/mL. Of those, 4% (34) underwent a repeat PSA test and 5% (49) underwent a mpMRI Prostate - 47% (23) were classified as PI-RADS 3 or greater. Four patients with a PI-RADS score of 1 or 2 were referred to Urology based on age or worrisome PSA trend. There were 5 patients referred to Urology without an upfront mpMRI due to age, initial degree PSA elevation, or a combination thereof. There were 4 patients who had a PI-RADS 3 or greater lesion identified on MRI with a pending recommendation. There were 17 patients with an initial elevated PSA who had a pending work-up (repeat PSA or mpMRI Prostate) at the end of the review period. In total, 28 patients were referred to Urology and 12 underwent a biopsy by the end of the review period. Prostate cancer was identified in 9 (75%) biopsies and 8 of those patients had further treatment or surveillance (1 patient lost to follow-up) (Table 1). Based on PSA trend or mpMRI Prostate results, 26 patients avoided a referral to urology after initial elevated PSA. Excluding the 17 patients who had pending work-ups, 48% (26 of 54) of patients with a complete work-up avoided referral to Urology (Fig. 2).

MCI Program (June 1, 2022-December 31, 2024)

From June 1, 2022 to December 31, 2024, a total of 1773 patients were screened. The percentage of patients who had a repeat PSA increased to 7.3% (130) while the percentage undergoing mpMRI Prostate was 5.2% (92) - 51% (47) were graded with a PI-RADS 3 lesion or greater and 53% (25) of those with a lesion had biopsy confirmed prostate cancer. In total, 50 patients were referred to Urology and 42% (21) of them had received definitive treatment at the time of publication (Table 2).

Adherence to AUA Guidelines

The AUA guidelines emphasize shared decision-making between clinicians and patients regarding prostate cancer screening. For men at average risk, PSA screening may begin between ages 45 to 50. For men with risk factors for prostate cancer - Black ancestry, genetic mutations, or a strong family history of prostate cancer – may be offered screening

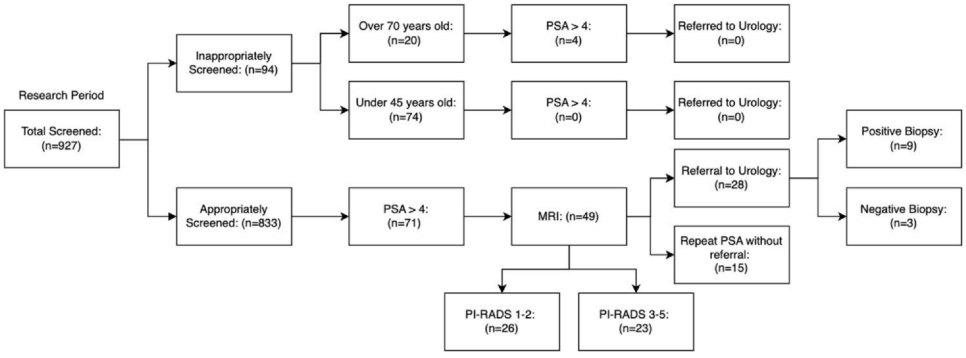


Figure 2. Flowchart of PSA screening, MRI evaluation, and urology referrals during research period (6/1/22-8/31/23). PSA, prostate-specific antigen.

Table 2
PSA results, mpMRI results and clinical outcomes of the MCI program (6/1/22-12/31/24).

Clinical Variables	Total Cohort (N = 1773)
Annual follow-up PSA	652 (36.8%)
Repeat PSA	130 (7.3%)
mpMRI performed	92 (5.2%)
PI-RADS 1	22 (1.2%)
PI-RADS 2	23 (1.3%)
PI-RADS 3	15 (0.8%)
PI-RADS 4	12 (0.7%)
PI-RADS 5	20 (1.1%)
Prostate biopsies	33 (1.9%)
Positive findings on biopsy	25 (1.4%)
Referrals to Urology	50 (2.8%)
Treatment for prostate cancer*	21 (1.2%)
Active surveillance	5 (0.3%)
RALP	11 (0.6%)
XRT	3 (0.2%)
Metastatic disease treatment	2 (0.1%)

RALP, robotic assisted laparoscopic prostatectomy; XRT, external beam radiation therapy.

* Four patients were lost to follow-up prior to definitive treatment.

starting at age 40. The AUA recommends a confirmatory PSA before proceeding to further diagnostic measures. Regular screening is advised every 2 to 4 years for men aged 50 to 69.

Screening for Patients Under 45

In total, 101 patients under 45 were screened. Of those, 74 were not appropriately screened according to AUA guidelines and 27 men were appropriately screened based on elevated risk factors for prostate cancer – 11 had a family history of prostate cancer, 16 identified as African American and 4 presented with both risk factors. No patients screened had a PSA value > 4 ng/mL (range 0.26-2.66 ng/mL for appropriately screened, 0.19-3.55 for not appropriately screened).

Screening for Patients Over 70

All 20 patients over the age of 70 who were screened were considered to not be guideline adherent. Within this group, 4 men presented with elevated PSA levels and 2 reported a family history of prostate cancer (PSA range 0.64-9.11 ng/mL). One patient had a mpMRI of the prostate without a significant finding and was recommended to follow-up for repeat testing. One patient had a repeat PSA that was < 4 ng/mL. One patient was recommended to avoid further work-up based on advanced age and one patient had an mpMRI prostate pending at the end of the review period.

Program Costs

The recent cost projection from the finance department for the prostate cancer screening program estimates an annual cost of $113,856 to serve approximately 1000 patients. This projection includes personnel costs, accounting for 20% effort from three employees, alongside expenses for mobile services ($1590 monthly), supplies ($325 monthly), clinical services ($3181 monthly) and PSA testing ($18.39/test). The total estimated cost for the program is $9488 per month.

DISCUSSION

The results of this study demonstrate the feasibility and clinical utility of a primary care led community-based PSA screening program. By shifting the initial workup and triage including repeat PSA testing and mpMRI from the urologist's office to the primary care setting we identified clinically significant prostate cancer while maintaining 90% adherence to AUA guidelines. Notably our model allowed 48% of patients with an elevated initial PSA to safely avoid a urologic referral through normalized repeat testing or negative MRI findings. This suggests that the urologic gap in underserved populations can be narrowed without overwhelming existing specialist capacity.

A critical component of this model's feasibility is its economic structure. The program operates at a relatively low cost of approximately $10 to $15 per patient for the initial screening event which covers supplies and laboratory fees.[22] By utilizing primary care led triage the system avoids the significantly higher costs associated with immediate specialist consultation and biopsy for every patient with an initial PSA elevation. Furthermore, while mpMRI represents a higher upfront diagnostic cost the philanthropic funding model used here demonstrates how targeted investments in imaging can result in long term savings by reducing the volume of unnecessary biopsies and specialist visits costs that are often prohibitive for uninsured or underinsured populations in urologic deserts.

The disparities in prostate cancer access addressed by this program mirror systemic inequalities seen in other screening detectable malignancies. For instance, geographic and socioeconomic barriers in breast and colorectal cancer screening often lead to late-stage presentation in minority cohorts.[23,24] By utilizing a primary care led multidisciplinary model our program addresses the access gap in a manner similar to successful community-based mammography or FIT DNA initiatives. This indicates that the primary care office can effectively serve as a hub for complex oncologic triage in underserved regions provided there is a clear algorithm and institutional support for diagnostic imaging.

While national guidelines offer less certain direction for men outside the 45-69 age range, our primary care-led model successfully adapted to these clinical gray areas. By utilizing the collaborative partnership with urology, the family medicine physician was able to apply nuanced risk stratification to the < 45 and > 70 cohorts, ensuring that screening remained both safe and purposeful for these outliers. This specific focus on the extremes of the screening age spectrum highlights the versatility of the triage algorithm when managed by a primary care provider who is well-versed in the collaborative protocol.

This study has several limitations. First, its retrospective design and the presence of pending workups for a subset of the cohort (18.5% of those with elevated PSA) may affect the final referral and cancer detection rates reported. Second, the single-center, non-profit nature of the program may limit the generalizability of this model to traditional "fee-for-service" systems where imaging costs are a barrier for the uninsured. Third, loss to follow-up remains a challenge in underserved populations, as seen in the participants who did not complete biopsy or definitive treatment. Finally, as a feasibility study, we lacked a control group to directly compare the efficiency of this model against a standard urology-referral pathway.

CONCLUSION

This community-based, primary care–led screening model demonstrates that structured PSA screening can effectively reach underserved populations while maintaining high guideline adherence. By integrating repeat PSA testing and mpMRI triage, we significantly reduced unnecessary urologic referrals and identified clinically significant disease in a high-risk cohort. This framework provides a scalable, replicable strategy to bridge urologic care gaps and mitigate health disparities without overextending specialist resources.

Disclosures

There were no external sources of funds supporting this work.

Ethical Declaration

This study was reviewed and approved by the Institutional Review Board at the Moncrief Cancer Institute (STU-2023-0594).

CRediT Authorship Contribution Statement

Caleb Q. Ashbrook: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Thomas C. Troia**: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Solomon L. Woldu**: Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization. **Keith Argenbright**: Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors have no conflict of interest to declare.

Appendix A. Supporting Information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.urology.2026.04.002.

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